

39. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator given by $T(x, y) = (x + 2y, -3x)$.

Let B be the following basis B of $\mathbb{R}^2$: $B = \{(0, 1), (-1, 2)\}$.

Let $v = (1, 2)$.

(a) Compute $[T]_B$ in two ways:

1. Apply directly the definition of $[T]_B$.
2. First find $[T]_E$ (where E is the standard basis of $\mathbb{R}^2$), then apply change-of-basis.

(b) Compute $[T(v)]_B$ in two ways:

1. First compute $[v]_B$ directly by the definition of a coordinate vector, then apply the theorem $[T]_B[v]_B = [T(v)]_B$.
2. First find $[T(v)]_E$, then apply change-of-basis.

40. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator whose matrix with respect to the basis $B = \{(1, 2), (1, 3)\}$ is $[T]_B = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$.

Use change-of-basis to determine an explicit formula for $T(x, y)$.

41. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation given by $T(x, y, z) = (x - 2z, y - 3z)$.

(a) Compute $[T]_B^C$ with respect to the bases $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ and $C = \{(1, 0), (1, 1)\}$ in two ways:

1. Apply directly the definition of $[T]_B^C$.
2. First find the matrix of T with respect to the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$, then apply change-of-basis.

(b) Let $v = (x, y, z) \in \mathbb{R}^3$. Compute $[v]_B$ and $[T(v)]_C$ directly by the definition of a coordinate vector, and verify that $[T]_B^C[v]_B = [T(v)]_C$.

42. Let $T: \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ be the linear operator given by $T(X) = AX - XA$, where $A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$.

(a) Compute $[T]_E$ with respect to the basis $E = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

(the standard basis of $\mathbb{R}^{2 \times 2}$).

(b) Use $[T]_E$ to find a basis and to determine the dimension of $\text{Ker}(T)$ and of $\text{Im}(T)$.